MIL-P-11859G
12 February 1985
SUPERSEDING
MIL-P-11859F
14 May 1976

MILITARY SPECIFICATION

PONTON BOATS, ALUMINUM, FOR FLOATING FOOTBRIDGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aluminum ponton boats used to support floating aluminum footbridges and light vehicular bridges.

1.2 Classification. The ponton boats shall be the following types, as specified (see 6.2):

Type I - For supporting floating aluminum footbridges.
Type II - For supporting floating aluminum footbridges and light vehicular bridges.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-21929	- Plastic Material, Cellular Polyurethane, Foam-In-Place, Rigid, (2 and 4 Pounds Per Cubic Foot).
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13211E8580	- Ponton Boat, Welded Aluminum, for Floating Footbridge.
TA13211E8570	- Ponton Boat, Welded Aluminum for Floating Footbridge (with Light Vehicular Capabilities).
D13207E1366-1	- Packing for: Ponton Boat, Aluminum for Floating Footbridge.
D13207E1366-2	- Packing for Floating Footbridge: Ponton Crate Details, Level A.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

ANSI/AWS A5.10 - Aluminum and Aluminum Alloy Bare Welding Rods and Electrodes.

ANSI/AWS D1.1 - Structural Welding Code - Steel, Section 5, Qualification.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 2341 - Rigid Urethane Foam.

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The ponton boats shall be in accordance with Top Assemblies TA13211E8580 for Type I and TA13211E8570 for Type II, and as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the ponton. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative. The dimensions shown on the drawings apply at temperatures from 68° F to 70° F.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The ponton boats shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the ponton boats may be newly fabricated from recovered materials to the maximum extent practicable, provided the ponton boats produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the ponton boats.

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3.3.3 Polyurathene foam. The space between the floor sheet and the skin shall be filled utilizing a polyurethane froth foam-in-place process. The floor sheet, skins, and structural members in the cavity shall be cleaned in accordance with MIL-T-704 prior to filling. The material as foamed and in place shall conform to MIL-P-21929, class 1, and to type 3416756660 material as specified in ASTM D 2341.

3.3.3.1 Preparation of sample blocks. Sample blocks necessary for tests shall be prepared as specified in MIL-P-21929.

3.3.3.2 Foaming procedure. The material shall be dispensed as a froth into the ponton boat through an opening in the stern sheet. The ponton boat shall be restrained to prevent shape distortion from the force of the expanding material. When installed, the plastic material shall show sufficient expansion to seal off all openings leading to the floatation compartment (see 4.5.2.1). Excess material shall be trimmed flush to the adjacent metal prior to painting the ponton boat.

3.3.3.3 Foaming process. Prior to the foaming of any ponton boat, the contractor shall prepare and submit to the contracting officer for approval, a written foaming procedure defining process and control. This procedure shall include, but not be limited to, the factors listed below. Process changes affecting any of the factors shall be approved by the contracting officer or designated representative prior to adoption of the change. No change in material suppliers will be made without approval of the contracting officer or designated representative.

a. Method of restraining the ponton boat.

b. Temperature range of the ponton boats prior to foaming.

c. Temperature range of foam components.

d. Method of placing material in ponton boat.

e. Total foam dispensing time.

f. Calibration of equipment.

g. Ponton boat cure time in restraining fixture.

3.4 Identification markings. The ponton boat shall be identified in accordance with MIL-STD-130. The identification plates shall be as shown on the drawing and shall be attached in accordance with the drawing.

3.5 Finishing. The ponton boat shall be cleaned, treated, and painted in accordance with MIL-T-704, type B.

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3.6 Workmanship.

3.6.1 Metal fabrication. The surfaces of all complete members used in fabrication shall be free from buckling, kinks, wrinkles, washboarding, burrs, scribe lines, and scratches. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Moving parts shall operate without binding.

3.6.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1, Structural Welding Code-Steel, Section 5, Qualification.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.6.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale and other foreign matter that can be removed by chipping and wire brushing. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the ponton boat to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch.

3.6.3.1 Aluminum welding. Aluminum welding shall be done either by the metal-inert-gas method or the tungsten-arc method. Filler metal shall conform to ANSI/AWS A5.10, classification ER4043. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed with a stainless steel wire brush or chipped but shall not be hammered. Preheating for welding is permissible providing the temperature does not exceed 400° F for a total time of 30 minutes. No process requiring the use of welding flux shall be used. Completed welds shall be free from discontinuties such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or splatter.

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article ponton boats shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article ponton boats shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of either test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

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4.4.3 Tests. Each ponton boat shall be tested as specified in 4.5.2.1. Failure of this test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The ponton boats shall be examined as specified herein for the following defects:

Major

101. Any part or components not in accordance with the QAP requirements as shown on the drawings.
102. Materials not as specified.
103. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatment.
106. Components missing, damaged, or not as specified.
107. Moving parts binding.
108. Welds missing.
109. Distortion of the floor sheet (washboarding).
110. Foaming process not in accordance with approved process.
111. Welding not in accordance with approved process.
112. Workmanship not as specified.

Minor

201. Any dimensions, other than those identified on the QAP requirements shown on the drawings, not as specified.
* 202. Weld defects such as base-metal cracks adjacent to or behind welds, crater cracks, or cracks in weld metal, spatter, under-cutting of base metal.
* 203. Casting containing blowholes, porosity, shrinkage defects, or cracks.
204. Presence of buckling, kinks, or scribe lines on complete members.
205. Sharp edges, slivers, burrs, or projections on finished surfaces.
206. Polyurethane filler material not filling the space between the inner floor and outer skin.
207. Cleaning, treating, and painting not as specified.
208. Identification marking not as specified.

* Examination for these items shall be made prior to painting.

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4.5.2 Tests.

4.5.2.1 Precautions. Since toxic isocyanate fumes may be evolved during the foaming operation, adequate ventilation shall be provided to reduce the isocyanate concentration in the air below 0.1 parts per million, or gas masks equipped with the appropriate gas filter shall be worn. Additional personnel protection shall consist of face shields, gloves and protective clothing.

4.5.2.2 Foaming process. The unfinished ponton shall be weighed, placed in the restraining fixture and foamed in accordance with the submitted foaming procedure (see 3.3.3.3). After the specified waiting period the ponton shall be removed from the fixture and reweighed to determine the amount of foam added. The increased weight of the ponton after foaming of more than 25 pounds or less than 15 pounds shall constitute failure of this test.

4.5.2.3 Foam block. The sample block (see 3.3.3.1) shall be examined for homogeneity, flame resistance, and oil resistance in accordance with MIL-P-21929. Nonconformance to MIL-P-21929 shall constitute failure of these tests.

4.6 Inspection of packaging.

4.6.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be a unit of product.

4.6.2 Examination. Each pack shall be examined for the defects listed below. Presence of one or more defects shall be cause for rejection.

113. Material not as specified for level A.
114. Construction not as specified for level A.
115. Marking illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Ponton boats shall be assembled in groups of 21 each and secured to the MIL-C-3774, type I modified crate base as shown on drawings 13207E1366-1 and -2. After the boats are secured to the crate base, the sides, ends, and top of the crate shall be positioned and secured.

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5.1.2 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.2 Marking.

5.2.1 Military marking. Ponton boats and shipping containers shall be marked in accordance with MIL-STD-129. The marking on the ponton boat shall include the National Stock Number, nomenclature, and contract number and shall be stenciled on the port side of the stern and on the starboard side forward, prior to the bow rise, approximately 3 inches below the gunwale.

5.2.2 Commercial marking. Commercial packs shall be marked in accordance with ASTM D 3951. Additionally, each shipping container shall be marked with the weight and cube.

6. NOTES

6.1 Intended use.

6.1.1 Type I ponton boat. The Type I ponton boat is intended for Army use in supporting floating footbridges.

6.1.2 Type II ponton boat. The type II ponton boat is intended for Marine Corps use in supporting floating footbridges and light vehicular bridges.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of ponton boat required (see 1.2).

c. When a first article is required for inspection and approval, and the number of units required (see 3.2).

d. Degree of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be an initial production item. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Quality assurance provisions (QAP) record. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

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"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 International standardization. Certain provisions of this specification are the subject of international standardization agreement ABCA-QSTAG-88. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

Custodians:
Army - ME

User activity:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-11859G	Ponton Boats, Aluminum, for Floating Footbridge

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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